

PROJECT: RED DESCENT

COMPREHENSIVE GAME DESIGN DOCUMENT (GDD) & PROTOTYPE SPEC

Objective for AI Developer: This document contains the complete structural, mechanical, and narrative design for a 2D procedural rogue-lite digging game. Your task is to use this specification to architect and build a working prototype, preferably using the **Godot Engine (GDScript)** due to its robust 2D tilemap capabilities.

1. High-Level Concept

Logline: A stranded zero-g asteroid miner crash-lands on Mars and must use ill-suited vacuum equipment to dig deep into the planet's hostile crust to scavenge parts for their ship, ultimately discovering an ancient human terraforming silo and sacrificing their rig to launch a dormant escape capsule to Earth.

- **Genre:** Action Rogue-lite / Digging Survival
- **Perspective:** 2D Side-scrolling / Grid-based Tilemap
- **Engine Preference:** Godot Engine (Targeting Desktop PC)
- **Visual Style:** Chunky, high-contrast pixel art. Martian reds/oranges contrasting with cold, battered space suit greys/blues.

2. Core Gameplay Loop

The game relies on a repetitive, addictive loop balancing risk and reward:

1. **The Dig (Run):** Player descends into procedurally generated Martian depths. They balance energy, drill heat, and environmental hazards to mine ores and ship parts.
2. **The Death / Return:** The run ends when the rig runs out of energy, overheats, or is crushed by the environment.
3. **The Wreckage (Hub):** Player spends gathered materials to smelt alloys, repair ship components, and permanently upgrade the rig's chassis, drill, and battery.
4. **Repeat:** Player launches again, able to dig deeper and survive longer.

3. Player Mechanics (The "Wrong Gear" Premise)

The protagonist is an asteroid miner whose gear is calibrated for zero-gravity vacuums, not high-gravity planetary crusts. This informs the entire moveset:

- **Drilling (Heat vs. Speed):** Digging generates heat. If the heat gauge maxes out, the rig takes hull damage. Different terrains affect this (e.g., Permafrost cools the drill, Basalt superheats it).
- **Micro-G Thrusters:** Replaces the standard jump/jetpack. Extremely weak in Martian gravity. Only useful for softening falls, short lateral dashes, or desperate bursts to escape cave-ins.
- **Kinetic Impactors (Bombs):** Designed to shatter loose space rock. On Mars, they cause massive, unpredictable chain-reaction collapses. High risk, high reward terrain clearing.
- **Tether Cables:** Zero-g anchoring harpoons repurposed as clumsy grappling hooks to pull the heavy rig out of deep vertical shafts.
- **Energy Management:** Every action drains the central battery. Moving through empty space drains slowly; active digging and thruster use drains it rapidly.

4. Environmental Hazards (The Enemy)

The environment is the primary antagonist. There are no traditional "monsters."

- **Cave-Ins:** Digging too wide or destroying bedrock causes the ceiling to collapse, dropping physics-based debris that damages the rig.
- **Dust Accumulation:** Fine Martian regolith builds up in the rig's gears. Requires finding cleaning stations or venting precious air to blast joints clean.
- **Subterranean Pockets:** Procedurally generated pockets containing:
 - *Toxic Gas:* Slowly corrodes hull integrity.
 - *Lava Tubes:* Massive ambient heat spikes.
 - *Radiation:* Scrambles the UI, map, and telemetry sensors.
- **Atmospheric Pressure:** The deeper the player goes, the higher the pressure/gravity density. The rig passively uses more energy and digs slower without late-game upgrades.

5. Procedural Generation Architecture

The underground map must feel organic but guarantee a playable path. The prototype must implement a layered procedural approach:

Algorithm Stack

1. **The Critical Path (Spelunky Method):** Divide the map into chunk grids. Carve a mathematical "guaranteed path" from top to bottom before generating terrain.
2. **Terrain Base (Perlin/Simplex Noise):** Fill chunks with continuous gradients to create organic sweeping veins of dirt, rock, and ores.
3. **Caverns (Cellular Automata):** Run "Game of Life" rules on empty blocks to carve out natural-looking cave systems and lava tubes.

Biome Shifts (Depth Layers)

Biome Name	Depth	Generation Style & Hazards
The Crust	0m - 500m	Heavy noise. Loose dirt, scattered rocks, frequent cave-ins. Warm reds/browns.
The Mantle	500m - 1000m	Dense basalt walls. Cellular automata caves. Lava tubes and gas pockets. Dark reds, glowing orange.
The Ruins	1000m+	Shift to rigid, chunk-based dungeon gen (90-degree angles). Indestructible architecture. Cold blues/silvers.

6. Progression & Upgrades

In-Run (Temporary)

Items found or crafted during a dive that are lost upon death:

- **Coolant Flushes:** Instantly vents heat for a massive digging sprint.
- **Seismic Scanners:** Pings the map to reveal ores/hazards in a radius.
- **Nanobot Patches:** Consumable hull repairs.

Meta-Progression (Permanent Hub Upgrades)

Ores are smelted into specific alloys (Iron for hull, Copper for nav, Isotopes for drive) to repair the crashed ship and the rig.

- **Chassis Upgrades:** Base integrity, battery capacity, cargo space.
- **Drill Bits:** e.g., Laser Auger (melts ice, weak to rock), Tungsten Crusher (smashes rock, overheats fast).
- **Telemetry Beacon:** Allows the player to start a run at a deeper checkpoint (e.g., 500m) instead of the surface.

7. The Climax & The Endgame

The third act shifts from standard digging to archeological exploration and a high-stakes escape.

- **The Discovery:** At the bottom of the map, the player breaches an ancient, planetary-scale terraforming silo built by ancient humanity. The visual language shifts to rigid tilemaps, rusted bulkheads, and neon lights.
- **The Paradigm Shift:** The drill is no longer just for rock; it's used to hack power receptacles, chew through rusted vault doors, and navigate massive vertical drops using the micro-g thrusters.
- **The Ultimate Sacrifice:** The escape vehicle is a dormant, retro-futuristic capsule. To launch it, the player must dock their beloved, fully-upgraded mining rig into the terminal.
- **The Final Sequence:** The player transfers rig upgrades/power to the capsule, permanently draining the rig. A 60-second lockdown survival phase begins as the silo collapses, forcing the player to watch their rig get crushed outside the capsule window before blasting off toward Earth.

Prototype Development Phasing (For AI Implementation)

To the AI Developer: Please build the prototype in the following phases to ensure structural stability.

1. **Phase 1: Grid & Movement.** Implement a 2D tilemap. Create the player controller (grid-based or free-moving 2D physics). Implement the Micro-G Thrusters and basic gravity.
2. **Phase 2: Digging System.** Implement block destruction. Add the "Heat vs. Speed" mechanic and UI gauges (Heat, Energy, Hull).
3. **Phase 3: Basic ProcGen.** Implement Perlin noise terrain generation mapping to 3 block types (Dirt, Rock, Ore).
4. **Phase 4: Game Loop.** Add a surface hub. Implement basic inventory (collecting ore) and the death state (return to surface, lose inventory, keep meta-currency).
5. **Phase 5: Physics Hazards.** Implement the falling block logic (Cave-ins) when supporting blocks are destroyed.

End of Specification.